

OLIVER BABINGTON ELLIS

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Experience

Nanook Energy Advisors

July 2025 – Present

Software Engineer | Python · Kubernetes · AWS

- Architected the greenfield cloud and data platform to support quantitative research and systematic strategy development
- Engineered scalable agent infrastructure and execution environments to support automated workflows and research processes
- Productionised quantitative models and dashboards into robust Kubernetes workloads, establishing automated scheduling, monitoring, and self-service deployment templates
- Engineered distributed, parallel backtesting pipelines and frameworks leveraging Argo Workflows
- Built the data storage and serving layer using DuckLake, TimescaleDB, and FastAPI, delivering market and alternative datasets to researchers and discretionary traders
- Deployed and maintained core platform services on EKS, including MLflow for experiment tracking, automated CI/CD pipelines, and observability

Acorn Insurance

April 2023 – June 2025

Software Engineer | Python(FastAPI) Microservices · Kubernetes · Azure

- Sole developer on data science team and the main point of contact between other dev/dev-ops teams and data scientists
- Designed and built applications for critical real-time business operations, driving price optimisation and enabling predictive analytics for claims cost and intervention probabilities.
- Aided the data scientists in the standardization of ML pipelines, delivering reusable objects for rapid development and automated retraining, complemented by robust data preparation and pipelines
- Met all requirements with almost 100% availability and response times under SLA with good margins and scalability
- Maintained a key focus on observability (LGTM stack) during app development, ensuring app performance and stability, delivering said performance/stability metrics and insights to stakeholders for informed, confident decision making
- Lead the migration of data ingestion jobs to a real time event driven architecture

Mastercard

March 2022 – April 2023

Software Engineer | Go / Python Microservices · Kubernetes · Azure

- Key actor in the deployment of ML anti-fraud application scoring over 99.9% of UK bank transactions
- Provided real time support to maintain downtime/service degradation within the requirement of our SLA
- Worked between the MLEs and data engineers to harmonize simultaneous development/deployments

KX - First Derivatives

October 2021 – March 2022

Software Engineer, Capital Markets Graduate Programme | Kdb+ · Q

- Lone Software Engineer Deployed to a Formula 1 Race team tasked with the creation of Race Ops Application
- Co authored Data Ingestion pipelines to meet strict requirements to ensure data was available to trackside engineers
- Co-delivered a Proof of Concept to the Renault Clio Cup, securing both a product sale and future contracted work
- Developed a strong self-taught foundation in Q/Kdb+, going beyond the standard grad course through personal projects including building tickerplants across TimescaleDB, Influx, and Prometheus

Education

Imperial College London

Graduated May 2021

Bachelor of Engineering in Computer Engineering (EIE), Upper Second-Class Honours

Technical Skills

Languages Advanced: Python (NumPy, Pandas, Fastapi...)

Languages Intermediate: SQL, Kdb+/q, Go

Languages Beginner: C++, C#, R

DevOps and Infra: Kubernetes, Docker, Helm, Terraform, Azure Pipelines, Git, lambda, Argo Workflows

Data Storage/Processing: Postgres, DeltaLake, Mongo, Redis, Kafka, Azure Service Bus, Fivetran, Kdb+, TimescaleDB

Cloud: AWS, Azure, Google Cloud Platform, Firebase, DataBricks

Observability: Prometheus, Grafana, Kibana, Loki, Tiempo

Please turn over

Profile

As a Computer Engineering graduate with a proven track record of academic excellence, including achieving 100% in multiple exams (mostly maths). My diverse experience, sitting in different roles across machine learning and development teams, has equipped me with a comprehensive appreciation of the entire software lifecycle, from development to deployment. This experience is further enriched by my hands-on knowledge in both development and operations, emphasizing my ability to deliver robust solutions while understanding the synergy within a tech team.

My professional journey is marked by a strong preference for collaborative, in-person work environments where I can contribute to and learn from my peers. I thrive in settings that require teamwork and actively seek opportunities to engage in dynamic group projects. This inclination towards collaboration is matched by my genuine and substantial passion for technology. I am constantly engaged in enhancing my technical knowledge, whether it be through mastering the fundamentals or exploring the bleeding edge of tech advancements. This enthusiasm extends beyond my professional life, by reading extensively and frequently attending tech events, immersing myself in the latest industry trends and connecting with fellow tech enthusiasts.

In summary, I am a driven and adaptable professional with a keen interest in all things tech. My approach combines a solid foundation in mathematics, a wide-ranging technical skill set, and a sincere commitment to continuous learning and professional growth.

Projects and Courses

Time Priority Order Book | *Python* **2024**

- Implemented order book to accept market, limit and stop orders for simulations of historic market events/exchanges

Recruitment AI web app | *React (Typescript), Python, Firebase* **November 2023**

- Leveraged OpenAI assistants to develop a user-friendly web app, enabling non-technical recruiters to interact effectively with candidates in technical screenings (GPT wrapper).
- Integrated drag-and-drop functionality for easy upload and AI-driven analysis of CVs and job descriptions, converting them from PDF to text for processing
- Established a robust testing framework to ensure the reliability of the web app, while continuously updating agents to align with the latest advancements in AI

Practical Time Series Analysis | *Python, r* **June 2024**

- 6 week online course supplied by The State University of New York
- Gained proficiency in time series analysis techniques including ARIMA, SARIMA, and exponential smoothing